

Assignment 1: Take sample dataset and apply all clustering algorithm studied in class.

```
import pandas as pd
import numpy as np
import datetime
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler,MinMaxScaler
from sklearn.metrics import silhouette_score
from sklearn.metrics import silhouette_samples
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans, AgglomerativeClustering, DBSCAN
import scipy.cluster.hierarchy as shc
from sklearn.neighbors import NearestNeighbors
import warnings
warnings.filterwarnings("ignore")
```

```
pd.set_option("display.precision", 2)
pd.options.display.float_format = '{:.2f}'.format
data=pd.read_csv("/content/Customer_Data.csv",parse_dates=['CustCreationDate
data.sample(5)
```

	Customer_ID	Customer_Status	Old_Class	Geographical_ID	CustCreat
14111	114112	Inactive	C	2	20
58787	158788	Inactive	A	22	20
74157	174158	Active	C	21	20
47192	147193	Active	A	19	20
29452	129453	Active	C	14	20

```
data.shape
```

```
(88810, 13)
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 88810 entries, 0 to 88809
Data columns (total 13 columns):
#   Column                                Non-Null Count  Dtype
---  -
#   Column                                Non-Null Count  Dtype
```

```

0    Customer_ID      88810 non-null int64
1    Customer_Status  88810 non-null object
2    Old_Class        88810 non-null object
3    Geographical_ID  88810 non-null int64
4    CustCreationDate  88810 non-null datetime64[ns]
5    Net_Purchase(IRR) 88810 non-null float64
6    Avg_Invoice_Row_No 88810 non-null float64
7    Waste_Rate        88810 non-null float64
8    Rejected_Rate      88810 non-null float64
9    Avg_Purchase_In_Month 88810 non-null int64
10   Date_Of_Last_Purchase 88591 non-null datetime64[ns]
11   Unnamed: 11        0 non-null float64
12   Unnamed: 12        0 non-null float64
dtypes: datetime64[ns](2), float64(6), int64(3), object(2)
memory usage: 8.8+ MB

```

```

drop_columns=["Customer_ID","Geographical_ID","Unnamed: 11","Unnamed: 12","O
data.drop(drop_columns,axis=1,inplace=True)
data.sample(5)

```

	Customer_Status	CustCreationDate	Net_Purchase(IRR)	Avg_Invoice_Row_No
52176	Active	2019-08-03	1614443105.00	14.8
11974	Active	2019-05-15	283111619.60	6.3
57050	Active	2019-08-08	433122354.60	10.0
55397	Inactive	2019-08-08	106424542.70	13.6
25109	Inactive	2019-06-22	0.00	0.0

```
data["Customer_Status"].value_counts()
```

	count
Customer_Status	
Active	68529
Inactive	20281

```
dtype: int64
```

```

data=data[data["Customer_Status"]=="Active"]
data.sample(2)

```

	Customer_Status	CustCreationDate	Net_Purchase(IRR)	Avg_Invoice_Row_No
9276	Active	2019-05-15	171131683.70	12.1
82059	Active	2021-01-19	192962095.40	12.7

```
data["Date_Of_Last_ Purchase"].isnull().sum()
```

```
np.int64(41)
```

```
print(data.shape)
data.dropna(inplace=True)
data.shape
```

```
(68529, 8)
(68488, 8)
```

```
data.describe().T
```

	count	mean	min	25%	75%
CustCreationDate	68488	2019-10-09 02:30:54.035743232	2019-02-03 00:00:00	2019-06-21 00:00:00	2019-12-01 00:00:00
Net_ Purchase(IRR)	68488.00	581019938.72	-209812225.70	117715881.85	290919938.72
Avg_Invoice_Row_No	68488.00	11.78	0.00	6.94	18.75
Waste_Rate	68488.00	0.17	-0.42	0.00	0.50
Rejected_Rate	68488.00	0.98	-3.19	0.00	3.19
Avg_Purchase_In_Month	68488.00	36.16	0.00	17.00	54.17
Date_Of_Last_ Purchase	68488	2021-12-22 07:53:31.026749184	2020-01-01 00:00:00	2022-01-08 00:00:00	2023-01-01 00:00:00

```
data=data[data["Net_ Purchase(IRR)"]>=0]
data=data[data["Waste_Rate"]>=0]
data=data[data["Rejected_Rate"]>=0]
data.shape
```

```
(68483, 8)
```

Create new features

```
data["age"] = data["CustCreationDate"].apply(lambda x : ((datetime.datetime.now()-x).days/365))
data["age"]
```

	age
0	2519
1	2517
2	2458
3	2450
5	2436
...	...
88805	1523
88806	1523
88807	1523
88808	1523
88809	1523

68483 rows × 1 columns

dtype: int64

```
data["recency"] = data["Date_Of_Last_ Purchase"].apply(lambda x : ((datetime
data["recency"]
```

```
recency
```

```
data.drop(["Date_Of_Last_ Purchase","CustCreationDate","Customer_Status"],ax:
```

```
data.sample(2)
```

3	2159	Net_	Avg_Invoice_Row_No	Waste_Rate	Rejected_Rate	Avg_P
5	2108	Purchase(IRR)				
74782	233083574.60		7.65	0.04	0.45	
...	...					
47104	496799754.10		14.63	0.01	0.04	
88805	1455					

```
data.shape
```

88807 1455

88808 1455

88809 1455

68483 rows × 1 columns

dtype: int64